Statistical and Machine Learning

Regularization and Bayesian method

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Regularization and Bayesian method

- Regularization
  - Recap: Linear regression
  - Ridge regression
  - Lasso
  - Elastic net, Group lasso and beyond
- Bayesian method
  - Recap: Random variable
  - Bayes' formula
  - Conjugate prior
  - Bayesian lasso and beyond

Regularization and Bayesian method

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Linear Regression

- Least squares linear regression $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$:

$$\hat{\beta} = \arg\min_{\beta} \underbrace{(\mathbf{Y} - \mathbf{X}\beta)^\top (\mathbf{Y} - \mathbf{X}\beta)}_{\sum_{i=1}^{n} \varepsilon_i^2}$$

Issues

- Multicollinearity among the covariates can make $\mathbf{X}^\top \mathbf{X}$ nearly singular, leading to numerical instability when inverting $\mathbf{X}^\top \mathbf{X}$, implausible coefficient estimates, and very large variances for $\hat{\beta}$.
- If $\mathbf{X}^\top \mathbf{X}$ is singular – for example, in $p > n$ settings – the OLS estimator does not exist in its usual form, because $\mathbf{X}^\top \mathbf{X}$ is not invertible.

Another look: Linear algebra

- If $p > n$, equation $\mathbf{Y} = \boldsymbol{X}\beta$ has infinite number of solutions.
  - All these lead to $RSS = 0$
  - We need constraints on these solutions!
- **Idea: Regularization**
  - Regularization adds a penalty term that restricts the set of admissible solutions.
  - It is closely related to the use of priors in Bayesian models.

Regularization

- Basic approach to regularization/penalization for OLS estimation

$$PLS(\beta) = (\mathbf{Y} - \mathbf{X}\beta)^\top(\mathbf{Y} - \mathbf{X}\beta) + \lambda pen(\beta)$$

$$\hat{\beta}_{PLS} = \arg\min_\beta PLS(\beta)$$

- In a generalized case, we can replace $(\mathbf{Y} - \mathbf{X}\beta)^\top(\mathbf{Y} - \mathbf{X}\beta)$ by twice the negative log-likelihood
- Here, λ is a tuning (or smoothing) parameter, and $pen(\beta)$ is a penalty function. The choice of λ is an important issue, discussed later.
- The following formulation can be viewed as an alternative:

$$\lambda \mapsto \hat{\beta}_{PLS}(\lambda)$$

Regularization

From optimization theory: for each λ there is a unique t, such that

$$\hat{\beta}_{PLS} = \arg\min_{\beta} (\mathbf{Y} - \mathbf{X}\beta)^\top (\mathbf{Y} - \mathbf{X}\beta) + \lambda pen(\beta)$$

$$\Updownarrow$$

$$\hat{\beta}_{PLS} = \arg\min_{\beta} (\mathbf{y} - \boldsymbol{X}\beta)^\top (\mathbf{y} - \boldsymbol{X}\beta)$$

$$\text{s.t.} \quad pen(\beta) \leq t$$

- The problem can also be interpreted as a constrained optimization problem. This leads to the shrinkage of $\hat{\beta}_{PLS}$, usually towards 0; larger penalties typically produce more shrinkage.
- Different choices of $pen(\beta)$ lead to different regularization methods.

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Ridge regression

- Classic approach of Andrey Nikolayevich Tikhonov (1943), Arthur E. Hoerl and Robert W. Kennard (1970)
- Idea: penalize the magnitude of parameter β via its squared Euclidean norm $\boldsymbol{\beta}^\top \boldsymbol{\beta} = ||\boldsymbol{\beta}||_2^2 =: pen(\boldsymbol{\beta})$

$$PLS(\boldsymbol{\beta}) = (\mathbf{y} - \boldsymbol{X}\beta)^\top(\mathbf{y} - \boldsymbol{X}\beta) + \lambda\boldsymbol{\beta}^\top\beta$$

- Extreme cases: let $\hat{\boldsymbol{\beta}}_{LS}$ and $\hat{\boldsymbol{\beta}}_{PLS}$ be OLS and ridge regression estimators. Then it follows that

$$\lambda \to 0 \Rightarrow \hat{\boldsymbol{\beta}}_{PLS} \to \hat{\boldsymbol{\beta}}_{LS}$$
$$\lambda \to \infty \Rightarrow \hat{\boldsymbol{\beta}}_{PLS} \to \mathbf{0}$$

Ridge regression

- The ridge estimator can be obtained as follows:

$$\hat{\boldsymbol{\beta}}_{PLS} = (\boldsymbol{X}^\top \boldsymbol{X} + \lambda \boldsymbol{I}_p)^{-1} \mathbf{X}^\top \mathbf{Y}$$

- For $\lambda > 0$, $\boldsymbol{X}^\top \boldsymbol{X} + \lambda \boldsymbol{I}_p$ is invertible even if $(\boldsymbol{X}^\top \boldsymbol{X})$ isn't. We can write

$$\begin{aligned}\hat{\beta}_{PLS} &= (\boldsymbol{X}^\top \boldsymbol{X} + \lambda \boldsymbol{I}_p)^{-1} (\boldsymbol{X}^\top \boldsymbol{X})(\boldsymbol{X}^\top \boldsymbol{X})^{-1} \mathbf{X}^\top \mathbf{Y} \\ &= (\boldsymbol{X}^\top \boldsymbol{X} + \lambda \boldsymbol{I}_p)^{-1} (\boldsymbol{X}^\top \boldsymbol{X}) \hat{\beta}_{LS}\end{aligned}$$

- We thus obtain $\hat{\boldsymbol{\beta}}_{PLS}$ from $\hat{\boldsymbol{\beta}}_{LS}$ by premultiplication by a "shrinking" matrix.
- For orthonormal covariates, $\boldsymbol{X}^\top \boldsymbol{X} = \boldsymbol{I}_p$, we have

$$|\hat{\boldsymbol{\beta}}_{j,PLS}(\lambda)| = \frac{1}{1+\lambda} |\hat{\boldsymbol{\beta}}_{j,LS}| \leq |\hat{\boldsymbol{\beta}}_{j,LS}|, \quad \text{for all} j$$

- Most coefficients will typically decrease in absolute value with increasing λ (so called shrinkage towards 0).

Bias-variance trade off

- For OLS estimation, under linearity, $E(\varepsilon) = \mathbf{0}$, $E(\varepsilon\varepsilon^T) = \sigma^2\mathbf{I}_n$:
  $E(\hat{\beta}_{LS}) = \beta$,
  $Var(\hat{\beta}_{LS}) = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}$
- For ridge regression, under the same assumptions:
  $E(\hat{\beta}_{PLS}) = (\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I}_p)^{-1}\mathbf{X}^\top\mathbf{X}\beta \neq \beta$,
  $Var(\hat{\beta}_{PLS}) = \sigma^2(\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I}_p)^{-1}\mathbf{X}^\top\mathbf{X}(\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I}_p)^{-1}$
- One can show that $Var(\hat{\beta}_{PLS}) \leq Var(\hat{\beta}_{LS})$
- In the matrix sense: $Var(\hat{\beta}_{LS}) - Var(\hat{\beta}_{PLS})$ is positive semi-definite.

Ridge regression – choice of λ

- Through shrinkage, the ridge estimator typically has lower variance but higher bias than the OLS estimator.
- Consider the mean squared error (MSE):

$$MSE = Bias^2 + Var$$

- The tuning parameter λ steers the balance between variance and bias. One can show that there is always a $\lambda > 0$ such that the MSE of $\hat{\beta}_{PLS}$ is smaller than that of $\hat{\beta}_{LS}$
- How should λ be determined? $\rightarrow$ **Cross-validation!**

Choice of λ through CV

1. Split the data set randomly into K groups of approximately equal size (for example $K = 5$ or $K = 10$)
2. Estimate $\hat{\beta}_{PLS,(-1)}(\lambda)$ on the last $K - 1$ data sets and evaluate the fit on the left-out first data set $D1$ of size $|D1|$:

$$CV(\lambda)_1 = \frac{1}{|D1|} \sum_{i \in D1} (Y_i - \mathbf{x}_i^\top \hat{\beta}_{PLS,(-1)}(\lambda))^2$$

3. Now the respective second, third, ... K-th data set is left out and the resulting $CV(\lambda)_k$ is calculated
4. Overall, construct the average over the $CV(\lambda)_k$

$$CV(\lambda) = \frac{1}{K} \Sigma_{k=1}^{K} CV(\lambda)_k$$

Choice of λ through CV – final thoughts

- Cross-validation provides a target function, $CV(\lambda)$, to be minimized with respect to the parameter λ.
- It aims to optimize predictive performance on new data, thereby reducing the risk of overfitting to the observed dataset.
- The concept can be extended to objective functions other than the least squares criterion.
- In logistic regression (and other generalized linear models), one typically uses the negative log-likelihood.
- A special case is $K = n$: leave-one-out cross-validation.

Ridge regression – summary

- l_2 penalization
- Closed-form solution
- Shrinkage and bias–variance trade-off
- Tuning parameter selected by cross-validation

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The Lasso

- LASSO is short for Least Absolute Shrinkage and Selection Operator. The penalization is defined by the penalty function

$$pen(\beta) := \Sigma_{k=1}^{p} |\beta_k| = ||\boldsymbol{\beta}||_1$$

- Replacing the squared 2-norm $||\boldsymbol{\beta}||_2^2$ in Ridge regresson by the 1-norm $||\boldsymbol{\beta}||_1$ (or $||\tilde{\boldsymbol{\beta}}||_1$) if $\boldsymbol{\beta} = (\beta_0, \tilde{\boldsymbol{\beta}})$ and β_0 is not penalized).

$$\boldsymbol{\beta}_{LASSO} := \arg\min_{\beta} (\mathbf{y} - \boldsymbol{X}\beta)^\top (\mathbf{y} - \boldsymbol{X}\beta) + \lambda \Sigma_{k=1}^{p} |\beta_k|$$

The Lasso

- We can formulate it as a constrained optimization problem

$$\beta_{LASSO} := \arg\min_{\beta} (\mathbf{y} - \boldsymbol{X}\beta)^{\top} (\mathbf{y} - \boldsymbol{X}\beta)$$

- We have one-to-one correspondence between λ and t

$$pen(\beta) = ||\beta||_1 \leq t$$

The Lasso

- Note: **the objective function is not differentiable**.
- We obtain the following estimating equation for the LASSO-estimator:

$$-2\mathbf{X}^\top(\mathbf{y} - \boldsymbol{X}\beta) + \lambda(0, sign(\beta_1), \ldots, sign(\beta_p))^\top = 0$$

- This equation has jump discontinuities at each $\beta_j = 0$ and can only be solved numerically.

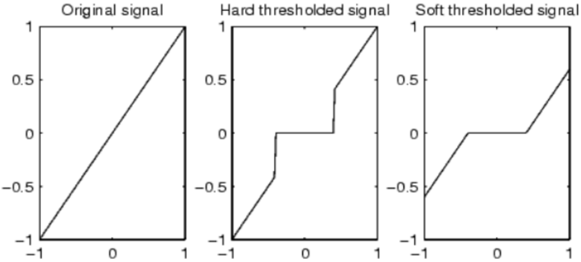
The Lasso: Variable selection

- For orthonormal covariates, $\mathbf{X}^\top \mathbf{X} = I_p$

$$\hat{\beta}_{j,Lasso}(\lambda) = sign(\hat{\beta}_{j,LS})(|\hat{\beta}_{j,LS}| - \tfrac{\lambda}{2})_+$$

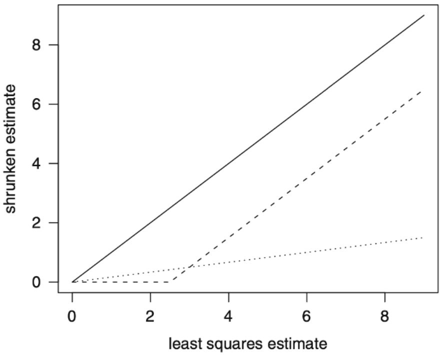
for all j, where $(z)_+ = max\{z, 0\}$.

This is also known as **soft thresholding**.



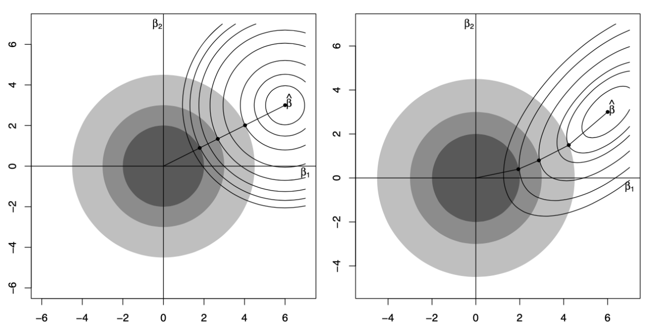
Lasso vs. Ridge regression

- Biased estimator
- Shrinkage
- Variable selection vs. close-form solution



Shrinkage effect for ridge (dotted), lasso (dashed) against least squares (solid) for $\lambda = 5$ and orthonormal covariates. While the LASSO-regularized estimate equals zero for a small coefficient, the ridge regression estimate is a scaled version of the OLS estimate with the of scaling determined by λ.
Source: Fahrmeir et al. (2013), Fig. 4.15

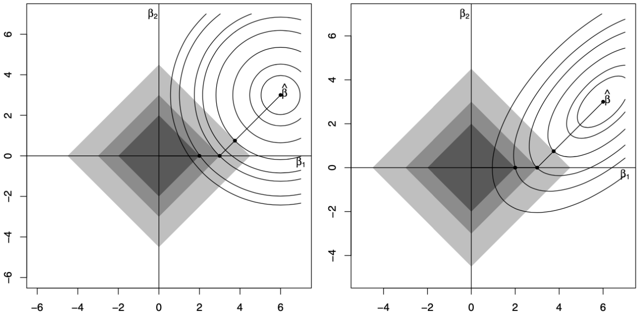
Lasso vs. Ridge regression



Thin lines correspond to contours of the LS criterion. Shaded areas correspond to different penalty values t. Connected dots correspond to the contrained estimator for different values of t.

Source: Fehrmeir et al. (2013), Fig. 4.12.

Lasso vs. Ridge regression



Thin lines correspond to contours of the LS criterion. Shaded areas correspond to different penalty values t. Connected dots correspond to the contrained estimator for different values of t.

Source: Fehrmeir, Lang, Kneib & Marx (2013), Fig. 4.12.

Lasso vs. Ridge regression

- The constrained optimum is achieved on the boundary of the circle/diamond with $pen(\beta) = t$, where a contour of the least squares criterion touches the circle/diamond.
- In the case of the LASSO, if the touching point is on a corner of the diamond, some of the coefficients are estimated to be exactly zero.
- Similarly as for ridge regression, we have

$$\lambda \to 0 \Rightarrow \hat{\boldsymbol{\beta}}_{PLS} \to \hat{\boldsymbol{\beta}}_{LS}$$
$$\lambda \to \infty \Rightarrow \hat{\boldsymbol{\beta}}_{PLS} \to \mathbf{0}$$

- We call the path of β with respect to λ a solution path.

Summary

- Regularization: shrinkage and bias-variance trade off
- l_2 regularization $\rightarrow$ shrinkage $\rightarrow$ biased but could achieve lower variance
- l_1 regularization $\rightarrow$ sparsity $\rightarrow$ variable selection $\rightarrow$ better interpretation
- Tunning the selection of parameters
- Extensions: regularized logistic regression/generalized linear regression, etc.

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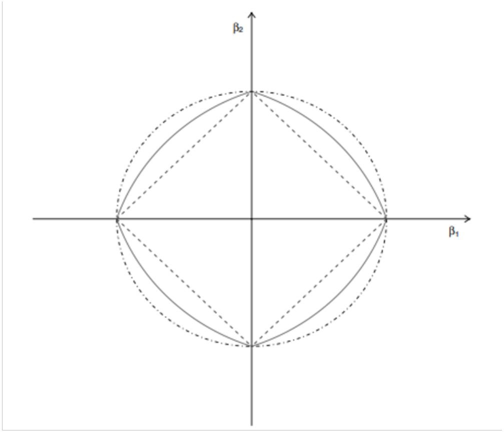
Elastic net (Hui Zou, JRSSB, 2005)

- Consider the extreme case $\mathbf{X} = (X_1, X_2, \ldots, X_p)$ with $X_1 = X_2$. Then the two covariates are indistinguishable, it is natural that they should enter or leave the model together.
- In this case:
  - for ridge regression, we always have $\hat{\beta}_1 = \hat{\beta}_2$
  - for the LASSO, the solution might be not unique

Elastic net

- Combination of l_1 and l_2 regularization, selected a group of correlated variables.
- Penalty: $\Sigma_{j=1}^{p}(\alpha|\beta_j| + (1-\alpha)\beta_j^2)$



Group LASSO (Ming Yuan, JRSSB, 2006)

- Suppose that, based on prior knowledge, we want to select a group of variables together.
- Consider the case $\mathbf{X} = (X_1, X_2, X_3)$ with $X_2 = X_1^2$. Then we may want to include X_1 and X_2 in the model simultaneously.
- This motivates adding group structure to the penalty.

Group LASSO

- Suppose we have G groups, and let $\boldsymbol{\beta}_{I_g}$ denote the subvector of coefficients corresponding to group I_g. For example, if the first group is $I_1 = \{1, 2\}$, containing variables (X_1, X_2), then

$$\boldsymbol{\beta}_{I_1} = (\beta_1, \beta_2).$$

- The group lasso penalty is

$$\sum_{g=1}^{G} \|\boldsymbol{\beta}_{I_g}\|_2.$$

- Note: this is not ordinary l_2 regularization. Instead, it is a sum of l_2-norms over groups.

More about regularization

- Sparse group lasso (Noah Simon et al., 2011, *JCGS*)

$$(1-\alpha)\lambda \sum_{l=1}^{m} \sqrt{p_l}\,\|\boldsymbol{\beta}^{(l)}\|_2 \;+\; \alpha\lambda\|\boldsymbol{\beta}\|_1$$

- (Latent) overlapping group lasso (Laurent Jacob et al., 2009, *ICML*)
- Fused lasso (Robert Tibshirani et al., 2005, *JRSSB*)
- ...

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Random variable

Are the following statements right or wrong?

a) If the probability of an event is zero, this event never happens
b) If the probability of an event is one, this event must happen

Random variable

- In general, both statements are **false**.
- Consider a random variable $X \sim U(0,1)$. For any single point $x \in [0,1]$, $P(X = x) = 0$. In particular, $P(X = 0.5) = 0$.
- However, this does **not** mean that the event $\{X = 0.5\}$ is impossible. It is a possible outcome; it simply has probability zero.
- Likewise, an event with probability one does not need to occur for every possible outcome. Such an event is said to happen **almost surely**, which means it may fail only on a set of probability zero.
- Usually, we cannot let a random variable be a number, instead, we describe random variables via their distribution.

Random variable

- Consider a random variable $X \sim N(\mu, \sigma^2)$
  - a) X is a random variable
  - b) Are μ and σ random variables or unknown parameters?

Random variable

- In **frequentist statistics**, μ and σ are treated as fixed but unknown parameters. In Bayesian statistics, they are treated as random variables with prior distributions.
- In the **frequentist framework**, the goal is to estimate unknown parameters of interest from the observed data.
- In the **Bayesian framework**, the goal is to use the observed data to update prior beliefs and obtain the posterior distribution of the parameter(s) of interest.

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Bayes' formula

The law of total probability

For two events A and B with $P(B) > 0$ the probability of A conditioning on B is given as:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} = \frac{P(B|A)P(A)}{P(B|A)P(A)+P(B|A^c)P(A^c)}$$

Bayes' theorem (normalized)

Let $p(\cdot)$ denote either a probability density function (for a continuous random variable) or a probability mass function (for a discrete random variable). Then,

$$p(\theta \mid y) = \frac{p(y \mid \theta)\, p(\theta)}{p(y)}.$$

Example: "The Boy Who Cried Wolf"

- "The tale concerns a shepherd boy who repeatedly tricks nearby villagers into thinking wolves are attacking his flock. When one actually does appear and the boy again calls for help, the villagers believe that it is another false alarm and the sheep are eaten by the wolf. In later English-language poetic versions of the fable, the wolf also eats the boy' ' (From Wiki)
- We give an illustration about this story. Let's assume at the beginning, the villagers believe that there's a 90% probability the boy is "completely honest", 5% he is "not trustworthy" and 5% he is "malicious".

Example: “The Boy Who Cried Wolf”

- Let’s consider the boy’s claim that there is a wolf. For different impressions of the boy, assume that

	Probability that the wolf actually exists (villagers believe the boy tells the truth)	Probability that there is no wolf (villagers believe the boy lies)
completely honest	99%	1%
not trustworthy	50%	50%
malicious	10%	90%

- What is the probability that the boy claims there is a wolf when there is actually no wolf?
  - $P(\text{no wolf}) = ??? = 0.079$
- After the villagers observe that the boy lied, what is the probability that they still believe the boy is completely honest?
  - $P(\text{completely honest} \mid \text{no wolf}) = ??? = 0.114$
- There is a 90% probability that the boy is “completely honest”, a 5% probability that he is “not trustworthy”, and a 5% probability that he is “malicious”.

Example: “The Boy Who Cried Wolf”

	Probability that the wolf actually exists (villagers believe the boy tells the truth)	Probability that there is no wolf (villagers believe the boy lies)
completely honest	99%	1%
not trustworthy	50%	50%
malicious	10%	90%

- After the villagers observed that the boy lied twice, what is the probability that they still believe the boy is completely honest?
  - $P(\text{completely honest} \mid \text{lied twice}) = \ldots = 0.0017$
- For the third time, the boy claims there is a wolf. What is the probability that the villagers believe there is a wolf?
  - $P(\text{wolf} \mid \text{lied twice}) = \ldots = 0.196$
    - “Once broken, real trust is hard to rebuild.”
- Q: How many times must the boy tell the truth for the probability that the villagers believe he is completely honest to become larger than 0.5?

Bayesian Inference

- Model parameters $\boldsymbol{\theta}$ are considered random variables with (unknown) distributions (reflecting our beliefs or uncertainty about $\boldsymbol{\theta}$)
- Before data collection: prior distribution (density) $p(\boldsymbol{\theta})$
- After observation of data y: posterior distribution (density) $p(\boldsymbol{\theta}|\mathbf{y})$
- Bayes' theorem gives us the means to calculate $p(\boldsymbol{\theta}|\mathbf{y})$

$$\underbrace{p(\boldsymbol{\theta}|\mathbf{y})}_{posterior} = \frac{p(\mathbf{y}|\boldsymbol{\theta}) \cdot p(\boldsymbol{\theta})}{p(\mathbf{y})} = \frac{p(\mathbf{y}|\boldsymbol{\theta}) \cdot p(\boldsymbol{\theta})}{\int p(y|\boldsymbol{\theta}) \cdot p(\boldsymbol{\theta}) d\boldsymbol{\theta}} \propto \underbrace{p(\mathbf{y}|\boldsymbol{\theta})}_{likelihood} \cdot \underbrace{p(\boldsymbol{\theta})}_{prior}$$

- Bayesian inference can be interpreted as a learning procedure where the observed data updates our prior knowledge about $\boldsymbol{\theta}$ to the posterior via the likelihood

Bayesian Inference

- The posterior distribution contains all information about the parameter provided by the data and the prior distribution.
- Point estimators can be obtained from the posterior, for example as the posterior mode, mean, or median.
- It allows an intuitive interpretation: probability statements about the parameter given the data.
- A main issue is the specification of the prior distribution (for example, what is an uninformative prior?) $\rightarrow$ see also Statistical Inference 1 and 2.

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Bayesian inference

Bayes' theorem (normalized)

Let $p(\cdot)$ denote either a probability density function (for a continuous random variable) or a probability mass function (for a discrete random variable). Then,

$$p(\theta \mid y) = \frac{p(y \mid \theta)\, p(\theta)}{p(y)}.$$

Bayes' theorem (non-normalized)

$$p(\theta \mid y) \propto p(y \mid \theta)\, p(\theta)$$

posterior $\propto$ likelihood $\times$ prior

Conjugate prior – example

Suppose σ^2 is fixed and known, and $X_1, \ldots, X_n \mid \mu \sim N(\mu, \sigma^2)$ i.i.d., $\mu \sim N(\mu_{\text{prior}}, \sigma^2_{\text{prior}})$. We derive the posterior $p(\mu \mid x_1, \ldots, x_n)$ step by step.

Step 1: Write down the prior and the likelihood

Prior: $p(\mu) \propto \exp\left(-\frac{(\mu-\mu_{\text{prior}})^2}{2\sigma^2_{\text{prior}}}\right)$

Likelihood (product of n independent Gaussian densities):

$$
\begin{aligned}
p(x_1, \ldots, x_n \mid \mu) &= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right) \\
&\propto \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i-\mu)^2\right)
\end{aligned}
$$

Step 2: Form the unnormalised posterior

By Bayes' theorem, $p(\mu \mid x_1, \ldots, x_n) \;\propto\; p(x_1, \ldots, x_n \mid \mu) \cdot p(\mu)$.
Taking logs (and dropping constants that do not depend on μ):

$$\log p(\mu \mid x_1, \ldots, x_n) \;\propto\; -\frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2 - \frac{(\mu - \mu_{\text{prior}})^2}{2\sigma_{\text{prior}}^2}$$

Step 3: Expand the quadratic terms

Expand each square: $(\mu - \mu_{\text{prior}})^2 = \mu^2 - 2\mu_{\text{prior}}\mu + \mu_{\text{prior}}^2$

$$\sum_{i=1}^{n} (x_i - \mu)^2 = \sum_{i=1}^{n} x_i^2 - 2\mu \sum_{i=1}^{n} x_i + n\mu^2 = \sum_{i=1}^{n} x_i^2 - 2n\bar{x}\mu + n\mu^2$$

Substituting and dropping terms that do not depend on μ:

$$\log p(\mu \mid \mathbf{x}) \;\propto\; -\frac{1}{2\sigma^2}\left(n\mu^2 - 2n\bar{x}\mu\right) - \frac{1}{2\sigma_{\text{prior}}^2}\left(\mu^2 - 2\mu_{\text{prior}}\mu\right)$$

Step 4: Collect terms in μ^2 and μ

Define the **precision** (inverse variance) of each source:

$$a = \frac{1}{\sigma^2_{\text{prior}}} \quad \text{(prior precision)}, \qquad b = \frac{n}{\sigma^2} \quad \text{(data precision)}$$

Then: $\log p(\mu \mid \mathbf{x}) \;\propto\; -\frac{1}{2}(a+b)\mu^2 + (a\,\mu_{\text{prior}} + b\,\bar{x})\,\mu$

Step 5: Complete the square

A general quadratic $-\frac{A}{2}\mu^2 + B\mu$ can be written as
$-\frac{A}{2}\left(\mu - \frac{B}{A}\right)^2 + \frac{B^2}{2A}$
The last term does not depend on μ and can be absorbed into the proportionality constant.
Applying this with $A = a + b$ and $B = a\mu_{\text{prior}} + b\bar{x}$:

$$\log p(\mu \mid \mathbf{x}) \;\propto\; -\frac{a+b}{2}\left(\mu - \frac{a\,\mu_{\text{prior}} + b\,\bar{x}}{a+b}\right)^2$$

Step 6: Read off the posterior distribution

The expression above is the log-kernel of a Gaussian. Therefore:

$$\boxed{\mu \mid x_1, \ldots, x_n \;\sim\; N\Big(\mu_{\text{post}},\, \sigma^2_{\text{post}}\Big)}$$

with

$$\mu_{\text{post}} = \frac{a\,\mu_{\text{prior}} + b\,\bar{x}}{a + b} = \frac{\dfrac{\mu_{\text{prior}}}{\sigma^2_{\text{prior}}} + \dfrac{n\bar{x}}{\sigma^2}}{\dfrac{1}{\sigma^2_{\text{prior}}} + \dfrac{n}{\sigma^2}}$$

$$\sigma^2_{\text{post}} = \frac{1}{a + b} = \frac{1}{\dfrac{1}{\sigma^2_{\text{prior}}} + \dfrac{n}{\sigma^2}}$$

Interpretation

Posterior mean is a precision-weighted average of prior mean and sample mean:

$$\mu_{\text{post}} = \frac{a}{a+b}\,\mu_{\text{prior}} + \frac{b}{a+b}\,\bar{x}, \qquad a = \frac{1}{\sigma^2_{\text{prior}}}, \quad b = \frac{n}{\sigma^2}$$

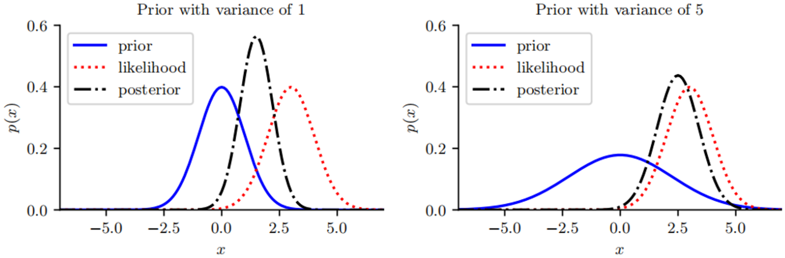
Posterior precision is additive: $\dfrac{1}{\sigma^2_{\text{post}}} = \dfrac{1}{\sigma^2_{\text{prior}}} + \dfrac{n}{\sigma^2}$, so σ^2_{post} shrinks with each new observation.

Limiting cases:

- $\sigma^2_{\text{prior}} \to \infty$: flat prior $\Rightarrow \mu_{\text{post}} \to \bar{x}$ (data dominate; posterior $=$ MLE sampling distribution)
- $n \to \infty$: $\mu_{\text{post}} \to \bar{x}$ (data always overwhelm prior asymptotically)
- $\sigma^2_{\text{prior}} \to 0$: tight prior $\Rightarrow \mu_{\text{post}} \to \mu_{\text{prior}}$ (prior dominates)

Conjugate pair: prior and posterior are both Gaussian

Gaussian-Gaussian prior



(a) *(b)*

Conjugate prior

- Prior and posterior both follow a normal distribution
- We say that a prior $p(\boldsymbol{\theta}) \in F$ is a conjugate prior for a likelihood function $p(D \mid \boldsymbol{\theta})$ if the posterior belongs to the same parametric family as the prior, that is, $p(\boldsymbol{\theta} \mid D) \in F$
- We then only need to update the parameters of the distribution, which has many computational benefits

Conjugate prior pairs

Likelihood	Prior	Posterior
Binomial	Beta	Beta
Negative Binomial	Beta	Beta
Poisson	Gamma	Gamma
Geometric	Beta	Beta
Exponential	Gamma	Gamma
Normal (mean unknown)	Normal	Normal
Normal (var. unknown)	Inverse Gamma	Inverse Gamma
Normal (mean, var. unknown)	Normal/Gamma	Normal/Gamma
Multinomial	Dirichlet	Dirichlet

Many sources for getting a list of congugate prioirs, see for example, this website.

Other priors

- Improper priors: if little or no prior information is available the prior distribution $p(\theta)$ should be rather flat.

$$p(\mu) \propto 1$$

- Jeffery's priors: let $\phi = f(\theta)$ be some invertible transformation of θ. We want to choose a prior that is invariant to this function f

$$p(\theta) \propto \sqrt{I(\theta)}$$

- Reference priors, invariant priors, hierachical priors

Regularization and Bayesian method

- Regularization
  - Recap: Linear regression
  - Ridge regression
  - Lasso
  - Elastic net, Group lasso and beyond
- Bayesian method
  - Recap: Random variable
  - Bayes' formula
  - Conjugate prior
  - Bayesian lasso and beyond

Bayesian linear regression

- Data: $(\mathbf{x}_i, y_i)$ $i = 1, \ldots, n$
- Model: $y_1, \ldots, y_n$ independent

$$y_i = \mathbf{x}_i^\top \beta + \epsilon_i$$

- Where $\beta = (\beta_1, \ldots, \beta_p)$ is the vector of regression coefficients.
- Normal prior

$$\mathbf{Y}|\boldsymbol{\beta} \sim N(\mathbf{X}\boldsymbol{\beta}, \boldsymbol{\Sigma})$$

$$\boldsymbol{\beta} \sim N(\boldsymbol{\beta}_o, \boldsymbol{\Sigma}_0)$$

Bayesian linear regression

- For a normal prior to β

$$\mathbf{Y}|\boldsymbol{\beta} \sim N(\mathbf{X}\boldsymbol{\beta}, \boldsymbol{\Sigma})$$
$$\boldsymbol{\beta} \sim N(\boldsymbol{\beta}_o, \boldsymbol{\Sigma}_0)$$

- The posterior distribution of β is a normal distribution with:

$$E(\beta|\mathbf{Y}) = \left(\mathbf{X}^\top\boldsymbol{\Sigma}^{-1}\mathbf{X} + \boldsymbol{\Sigma}_0^{-1}\right)^{-1}\left(\mathbf{X}^\top\boldsymbol{\Sigma}^{-1}\mathbf{Y} + \boldsymbol{\Sigma}_0^{-1}\boldsymbol{\beta}_0\right)$$
$$Var(\boldsymbol{\beta} \mid \mathbf{Y}) = \left(\mathbf{X}^\top\boldsymbol{\Sigma}^{-1}\mathbf{X} + \boldsymbol{\Sigma}_0^{-1}\right)^{-1}$$

- In particular, for the prior precision $\boldsymbol{\Sigma}_0^{-1} \to 0$ ("total prior uncertainty") and $\boldsymbol{\Sigma} = \sigma^2\mathbf{I}_n$

$$\boldsymbol{\beta}|\mathbf{Y} \sim N((\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}, \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1})$$

Bayesian Ridge Estimation

- For $\mathbf{\Sigma} = \sigma^2 \mathbf{I}_n$, $\mathbf{\Sigma}_0 = \tau^2 \mathbf{I}_p$ and $\beta_0 = \mathbf{0}$

$$\mathbf{Y}|\beta \sim N(\boldsymbol{X}\beta, \mathbf{\Sigma}) \ \beta \sim N(\beta_o, \mathbf{\Sigma}_0)$$

- The posterior distribution of β is the normal distribution with

$$E(\beta|\mathbf{Y}) = (\mathbf{X}^\top \mathbf{X} + \frac{\sigma^2}{\tau^2}\mathbf{I}_p)^{-1}\mathbf{X}^\top \mathbf{Y}$$

$$Var(\beta \mid \mathbf{Y}) = \sigma^2(\mathbf{X}^\top \mathbf{X} + \frac{\sigma^2}{\tau^2}\mathbf{I}_p)^{-1}$$

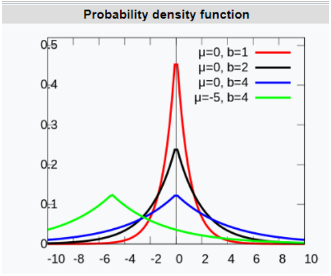
- The posterior mode (mean in this case) is equivalent to the Ridge-estimator with $\lambda = \frac{\sigma^2}{\tau^2}$. We can thus interpret the Ridge penalty on β as a normal prior, with the penalty parameter corresponding to the variance ratio $\lambda = \frac{\sigma^2}{\tau^2}$

Bayesian Lasso

The penalized least squares criterion for the Lasso can likewise be obtained with independent Laplace prior distributions

$$p(\beta_j) \propto exp(-\lambda|\beta_j|)$$

For either Ridge or Lasso, we can use the appropriate likelihood $p(y|\theta)$ instead of the least squares criterion in the generalized linear model case



$$f(x|\mu, b) = \frac{1}{2b} exp(-\frac{|x-\mu|}{b})$$

Bayesian Lasso and beyond

- For penalized regression, we have obtained values for λ using cross-validation.
- A Bayesian approach allows for simultaneous inference for β and the penalty amount measured through τ^2 by specifying additional (so-called hyper) priors for τ^2 and σ^2
- Fully Bayesian inference then uses simulation-based Markov chain Monte carlo (MCMC) inference
- Empirical Bayesian inference would estimate τ^2 and σ^2 from the data by maximizing the marginal likelihood obtained by integrating β out of the likelihood,

$$p(y|\tau^2, \sigma^2) = \int p(y|\beta, \sigma^2) p(\beta|\tau^2) d\beta$$

References

- ESLII: Chapters 3.4, 3.8
- PRML Chapters 3.1.4, 3.3
- PML: Chapters 2.3, 4.5, 4.6, 4.7, 11.3, 11.4, 11.7